WEEK -10

LOB LOGBOOK

NAME - LIKHITHA MAHADEVA

SID NUMBER - 2458682

ASSIGNMENT - WEEK 10

Model: "sequential_6"

Layer (type)	Output Shape	Param #
flatten_3 (Flatten)	(None, 784)	0
gaussian_noise_1 (GaussianNoise)	(None, 784)	0
dense_23 (Dense)	(None, 393)	308,505
dense_24 (Dense)	(None, 189)	74,466
dense_25 (Dense)	(None, 93)	17,670
dense_26 (Dense)	(None, 53)	4,982
dense_27 (Dense)	(None, 23)	1,242

Total params: 406,865 (1.55 MB)

Trainable params: 406,865 (1.55 MB)

Non-trainable params: 0 (0.00 B)

None

Model: "sequential_9"

Layer (type)	Output Shape	Param #
dense_33 (Dense)	(None, 53)	1,272
dense_34 (Dense)	(None, 93)	5,022
dense_35 (Dense)	(None, 189)	17,766
dense_36 (Dense)	(None, 393)	74,670
dense_37 (Dense)	(None, 784)	308,896
reshape_3 (Reshape)	(None, 28, 28)	0

Total params: 407,626 (1.55 MB)

Trainable params: 407,626 (1.55 MB)

Non-trainable params: 0 (0.00 B)

None

Model: "sequential_5"

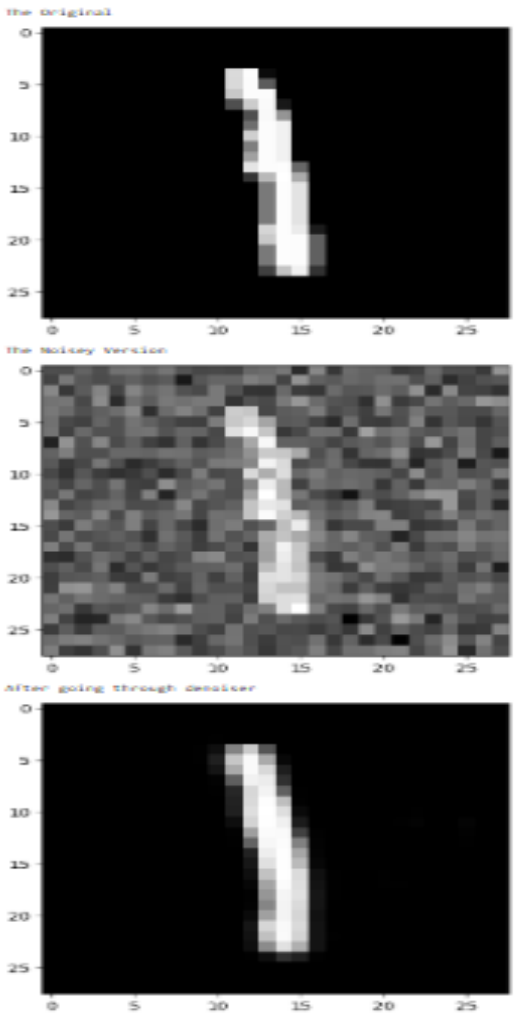
Layer (type)	Output Shape	Param #
sequential_3 (Sequential)	(None, 23)	406,865
sequential_4 (Sequential)	(None, 28, 28)	407,626

Total params: 814,491 (3.11 MB)

Trainable params: 814,491 (3.11 MB)

Non-trainable params: 0 (0.00 B)

None



Task-2

```
autoencoder.summary()
```

Model: "sequential_7"

Layer (type)	Output Shape	Param #
sequential_3 (Sequential)	(None, 91)	79,819
sequential_6 (Sequential)	(None, 36, 24, 1)	16,123

Total params: 95,942 (374.77 KB)

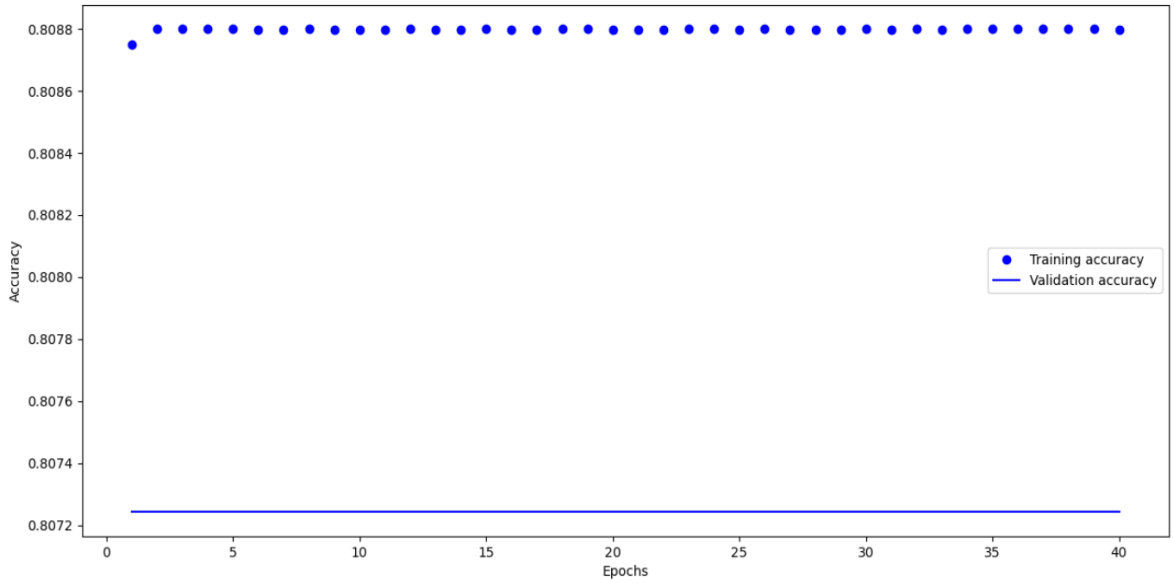
Trainable params: 95,942 (374.77 KB)

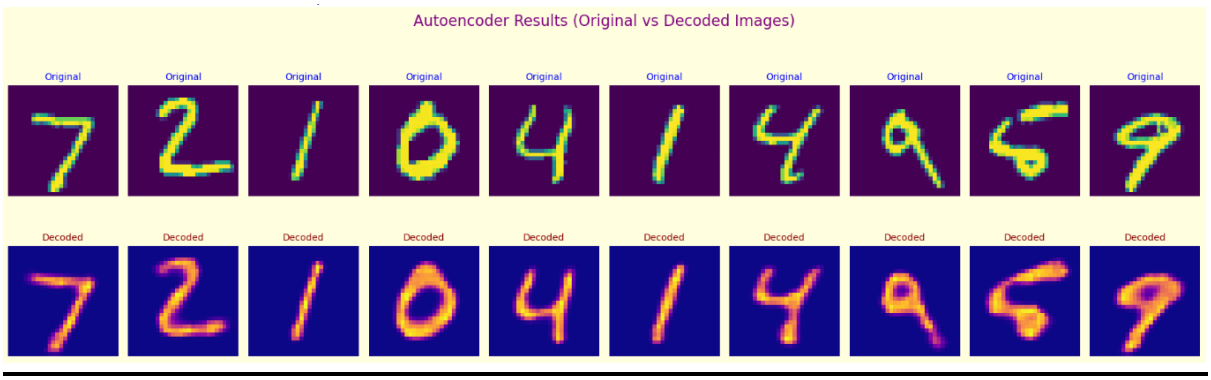
Non-trainable params: 0 (0.00 B)

```
history = autoencoder.fit(x=x_train,y=x_train,epochs=40,batch_size=128,shuffle=True,verbose=1,validation_data=(x_test, x_test))
```

```
Epoch 1/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 16s 26ms/step - accuracy: 0.8087 - loss: 0.0281 - val_accuracy: 0.8072 - val_loss: 0.0041
Epoch 2/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 12s 26ms/step - accuracy: 0.8088 - loss: 0.0040 - val_accuracy: 0.8072 - val_loss: 0.0040
Epoch 3/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 12s 26ms/step - accuracy: 0.8088 - loss: 0.0040 - val_accuracy: 0.8072 - val_loss: 0.0040
Epoch 4/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 12s 25ms/step - accuracy: 0.8088 - loss: 0.0040 - val_accuracy: 0.8072 - val_loss: 0.0040
Epoch 5/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 12s 25ms/step - accuracy: 0.8088 - loss: 0.0040 - val_accuracy: 0.8072 - val_loss: 0.0040
Epoch 6/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 12s 25ms/step - accuracy: 0.8088 - loss: 0.0040 - val_accuracy: 0.8072 - val_loss: 0.0040
Epoch 7/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 11s 24ms/step - accuracy: 0.8088 - loss: 0.0040 - val_accuracy: 0.8072 - val_loss: 0.0040
Epoch 8/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 12s 24ms/step - accuracy: 0.8088 - loss: 0.0040 - val_accuracy: 0.8072 - val_loss: 0.0040
Epoch 9/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 12s 25ms/step - accuracy: 0.8088 - loss: 0.0040 - val_accuracy: 0.8072 - val_loss: 0.0040
Epoch 10/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 12s 25ms/step - accuracy: 0.8088 - loss: 0.0040 - val_accuracy: 0.8072 - val_loss: 0.0040
Epoch 11/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 11s 24ms/step - accuracy: 0.8088 - loss: 0.0040 - val_accuracy: 0.8072 - val_loss: 0.0040
Epoch 12/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 12s 25ms/step - accuracy: 0.8088 - loss: 0.0040 - val_accuracy: 0.8072 - val_loss: 0.0040
Epoch 13/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 12s 26ms/step - accuracy: 0.8088 - loss: 0.0040 - val_accuracy: 0.8072 - val_loss: 0.0040
Epoch 14/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 12s 25ms/step - accuracy: 0.8088 - loss: 0.0040 - val_accuracy: 0.8072 - val_loss: 0.0040
Epoch 15/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 11s 24ms/step - accuracy: 0.8088 - loss: 0.0039 - val_accuracy: 0.8072 - val_loss: 0.0040
Epoch 16/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 11s 24ms/step - accuracy: 0.8088 - loss: 0.0038 - val_accuracy: 0.8072 - val_loss: 0.0038
Epoch 17/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 11s 22ms/step - accuracy: 0.8088 - loss: 0.0037 - val_accuracy: 0.8072 - val_loss: 0.0037
Epoch 18/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 12s 26ms/step - accuracy: 0.8088 - loss: 0.0037 - val_accuracy: 0.8072 - val_loss: 0.0037
Epoch 19/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 18s 22ms/step - accuracy: 0.8088 - loss: 0.0036 - val_accuracy: 0.8072 - val_loss: 0.0037
Epoch 20/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 13s 28ms/step - accuracy: 0.8088 - loss: 0.0036 - val_accuracy: 0.8072 - val_loss: 0.0037
```

```
Epoch 21/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 15s 31ms/step - accuracy: 0.8088 - loss: 0.0036 - val_accuracy: 0.8072 - val_loss: 0.0037
Epoch 22/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 13s 27ms/step - accuracy: 0.8088 - loss: 0.0036 - val_accuracy: 0.8072 - val_loss: 0.0036
Epoch 23/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 11s 24ms/step - accuracy: 0.8088 - loss: 0.0036 - val_accuracy: 0.8072 - val_loss: 0.0036
Epoch 24/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 12s 26ms/step - accuracy: 0.8088 - loss: 0.0036 - val_accuracy: 0.8072 - val_loss: 0.0036
Epoch 25/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 12s 25ms/step - accuracy: 0.8088 - loss: 0.0036 - val_accuracy: 0.8072 - val_loss: 0.0036
Epoch 26/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 11s 24ms/step - accuracy: 0.8088 - loss: 0.0036 - val_accuracy: 0.8072 - val_loss: 0.0036
Epoch 27/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 11s 23ms/step - accuracy: 0.8088 - loss: 0.0036 - val_accuracy: 0.8072 - val_loss: 0.0036
Epoch 28/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 11s 24ms/step - accuracy: 0.8088 - loss: 0.0036 - val_accuracy: 0.8072 - val_loss: 0.0036
Epoch 29/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 11s 24ms/step - accuracy: 0.8088 - loss: 0.0036 - val_accuracy: 0.8072 - val_loss: 0.0036
Epoch 30/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 11s 23ms/step - accuracy: 0.8088 - loss: 0.0036 - val_accuracy: 0.8072 - val_loss: 0.0036
Epoch 31/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 11s 24ms/step - accuracy: 0.8088 - loss: 0.0036 - val_accuracy: 0.8072 - val_loss: 0.0036
Epoch 32/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 11s 24ms/step - accuracy: 0.8088 - loss: 0.0036 - val_accuracy: 0.8072 - val_loss: 0.0036
Epoch 33/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 11s 23ms/step - accuracy: 0.8088 - loss: 0.0036 - val_accuracy: 0.8072 - val_loss: 0.0036
Epoch 34/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 11s 23ms/step - accuracy: 0.8088 - loss: 0.0036 - val_accuracy: 0.8072 - val_loss: 0.0036
Epoch 35/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 11s 24ms/step - accuracy: 0.8088 - loss: 0.0036 - val_accuracy: 0.8072 - val_loss: 0.0036
Epoch 36/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 11s 23ms/step - accuracy: 0.8088 - loss: 0.0036 - val_accuracy: 0.8072 - val_loss: 0.0036
Epoch 37/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 11s 23ms/step - accuracy: 0.8088 - loss: 0.0035 - val_accuracy: 0.8072 - val_loss: 0.0036
Epoch 38/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 12s 25ms/step - accuracy: 0.8088 - loss: 0.0035 - val_accuracy: 0.8072 - val_loss: 0.0036
Epoch 39/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 11s 24ms/step - accuracy: 0.8088 - loss: 0.0035 - val_accuracy: 0.8072 - val_loss: 0.0036
Epoch 40/40
469/469 ━━━━━━━━━━━━━━━━━━━━ 11s 23ms/step - accuracy: 0.8088 - loss: 0.0035 - val_accuracy: 0.8072 - val_loss: 0.0036
```




Autoencoder Results (Original vs Decoded Images)
Original
Original
Original
Original
Original
Original
Original
Original
Original
Original
Decoded
Decoded
Decoded
Decoded
Decoded
Decoded
Decoded
Decoded
Decoded
Decoded

WEEK -11

LOB LOGBOOK

NAME - LIKHITHA MAHADEVA

SID NUMBER - 2458682

ASSIGNMENT - WEEK 11

```
train(train_dataset,EPOCHS)
```

```
Time for epoch 41 is 20.717628717422485 sec
```

WEEK -9

LOB LOGBOOK

NAME - LIKHITHA MAHADEVA

SID NUMBER - 2458682

ASSIGNMENT - WEEK 9

```
es = EarlyStopping(monitor='val_loss', mode='min', patience=10, verbose=1)
mc = ModelCheckpoint('best_Trand-Flat_model_LSTM_GOLD.keras', monitor='val_loss', mode='min', verbose=1, save_best_only=True)
```

```
history = model.fit(X_train, y_train, batch_size=10, epochs=30,
                    validation_split=0.2, shuffle=True,
                    verbose=1, callbacks=[es, mc],
                    class_weight=class_weight_dict)
```

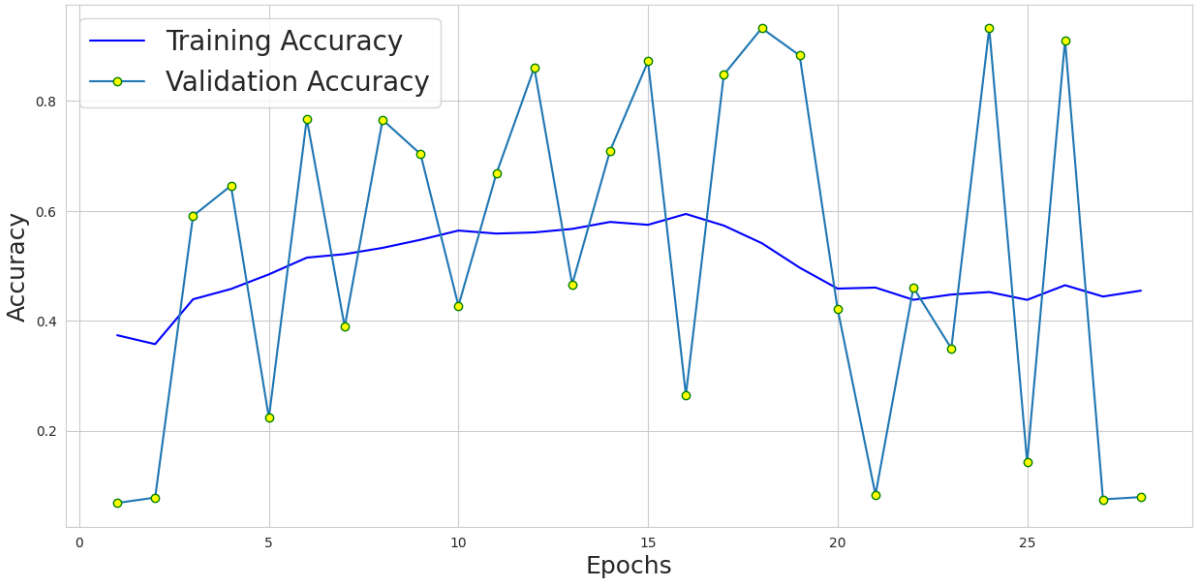
```
Epoch 1/30
856/861 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.5544 - loss: 0.6638
Epoch 1: val_loss did not improve from 0.50070
861/861 ━━━━━━━━━━━━━━━━━━━━ 8s 9ms/step - accuracy: 0.5632 - loss: 0.6564 - val_accuracy: 0.2427 - val_loss: 0.8692
Epoch 2/30
860/861 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.5763 - loss: 0.6520
Epoch 2: val_loss did not improve from 0.50070
861/861 ━━━━━━━━━━━━━━━━━━━━ 9s 11ms/step - accuracy: 0.5791 - loss: 0.6511 - val_accuracy: 0.4356 - val_loss: 0.7128
Epoch 3/30
855/861 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.5393 - loss: 0.6637
Epoch 3: val_loss did not improve from 0.50070
861/861 ━━━━━━━━━━━━━━━━━━━━ 10s 11ms/step - accuracy: 0.5482 - loss: 0.6523 - val_accuracy: 0.5472 - val_loss: 0.6576
Epoch 4/30
853/861 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.5902 - loss: 0.6258
Epoch 4: val_loss did not improve from 0.50070
861/861 ━━━━━━━━━━━━━━━━━━━━ 6s 6ms/step - accuracy: 0.5617 - loss: 0.6468 - val_accuracy: 0.7945 - val_loss: 0.5567
Epoch 5/30
853/861 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.5415 - loss: 0.6476
Epoch 5: val_loss did not improve from 0.50070
861/861 ━━━━━━━━━━━━━━━━━━━━ 6s 6ms/step - accuracy: 0.5327 - loss: 0.6511 - val_accuracy: 0.7815 - val_loss: 0.5747
Epoch 6/30
858/861 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.5393 - loss: 0.6581
Epoch 6: val_loss did not improve from 0.50070
861/861 ━━━━━━━━━━━━━━━━━━━━ 6s 7ms/step - accuracy: 0.5609 - loss: 0.6438 - val_accuracy: 0.3761 - val_loss: 0.7200
Epoch 7/30
858/861 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 0.5382 - loss: 0.6367
Epoch 7: val_loss did not improve from 0.50070
861/861 ━━━━━━━━━━━━━━━━━━━━ 11s 13ms/step - accuracy: 0.5460 - loss: 0.6436 - val_accuracy: 0.6918 - val_loss: 0.6141
Epoch 8/30
855/861 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.5514 - loss: 0.6456
Epoch 8: val_loss did not improve from 0.50070
861/861 ━━━━━━━━━━━━━━━━━━━━ 7s 8ms/step - accuracy: 0.5679 - loss: 0.6395 - val_accuracy: 0.4895 - val_loss: 0.6794
Epoch 9/30
853/861 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.5569 - loss: 0.6506
Epoch 9: val_loss did not improve from 0.50070
861/861 ━━━━━━━━━━━━━━━━━━━━ 6s 7ms/step - accuracy: 0.5114 - loss: 0.6862 - val_accuracy: 0.7875 - val_loss: 0.5579
Epoch 10/30
855/861 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.4510 - loss: 0.6742
Epoch 10: val_loss did not improve from 0.50070
861/861 ━━━━━━━━━━━━━━━━━━━━ 5s 6ms/step - accuracy: 0.4619 - loss: 0.6605 - val_accuracy: 0.5067 - val_loss: 0.6831
Epoch 10: early stopping
```

```python
error_metrics_report = classification_report(y_test, pred_binary)
print(error_metrics_report)
```

```
              precision    recall  f1-score   support

         0.0       0.89      1.00      0.94      1058
         1.0       0.00      0.00      0.00       137

    accuracy                           0.89      1195
   macro avg       0.44      0.50      0.47      1195
weighted avg       0.78      0.89      0.83      1195
```



WEEK -8

LOB LOGBOOK

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SID NUMBER - 2458682

ASSIGNMENT - WEEK 8

```
model = keras.Sequential([
    keras.layers.LSTM(10, activation='relu', input_shape=(50, 18)),
    keras.layers.Dense(2)

])
```

```
print(model.summary())
```

Model: "sequential_1"

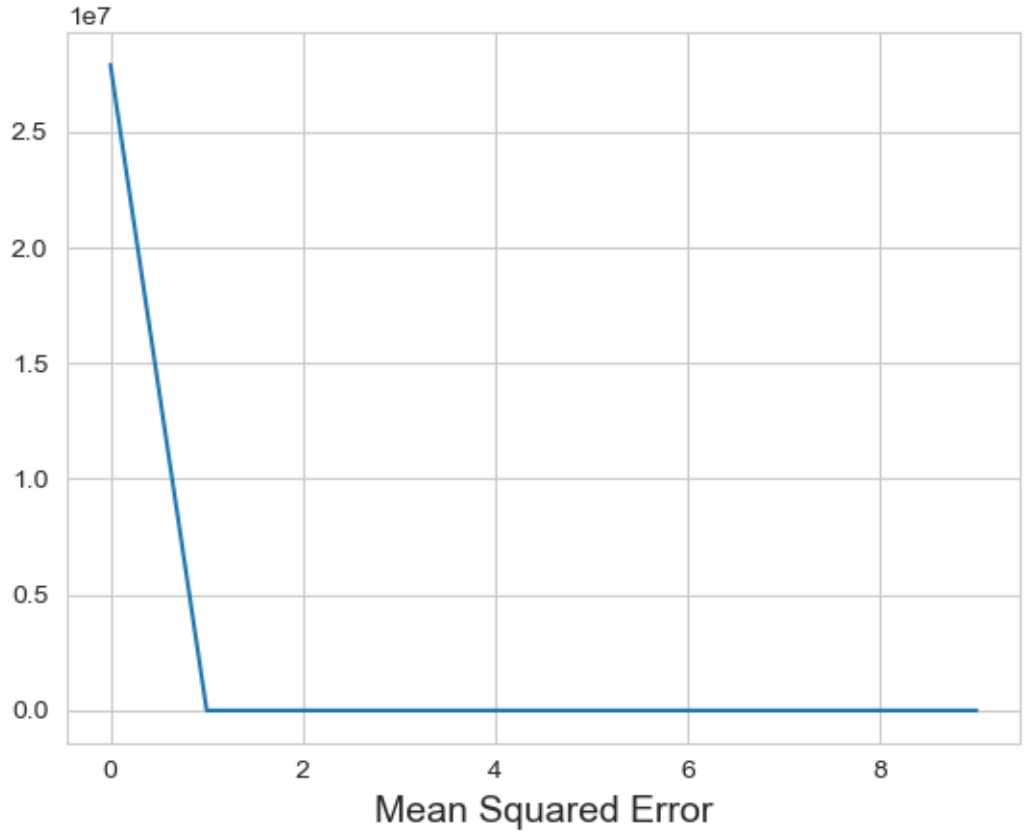
Layer (type)	Output Shape	Param #
lstm_1 (LSTM)	(None, 10)	1,160
dense_1 (Dense)	(None, 2)	22

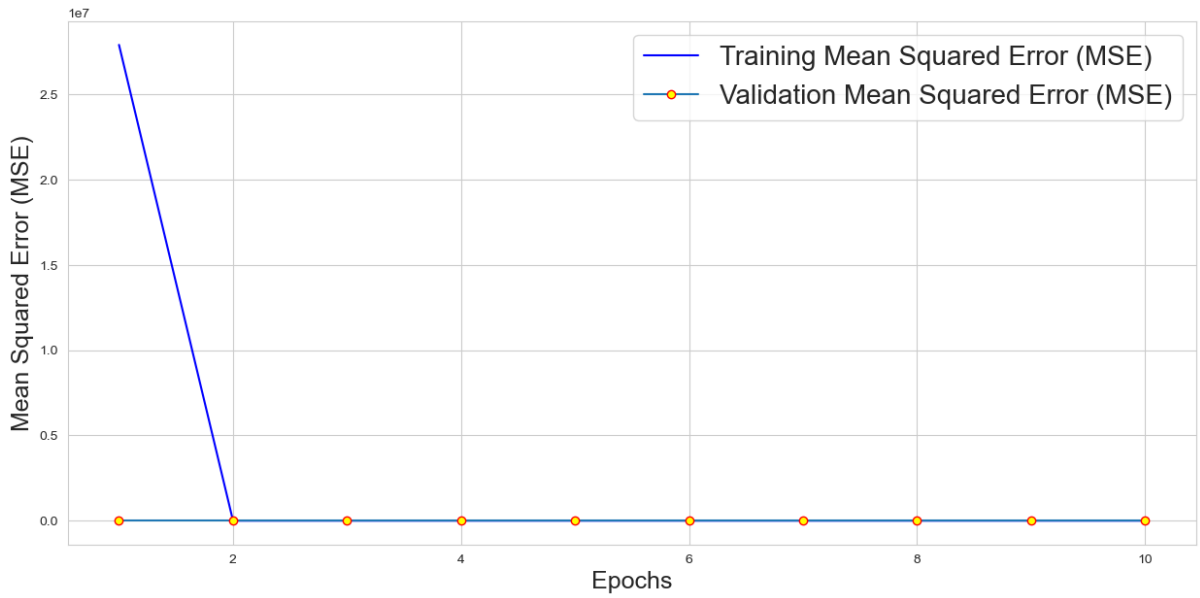
Total params: 1,182 (4.62 KB)

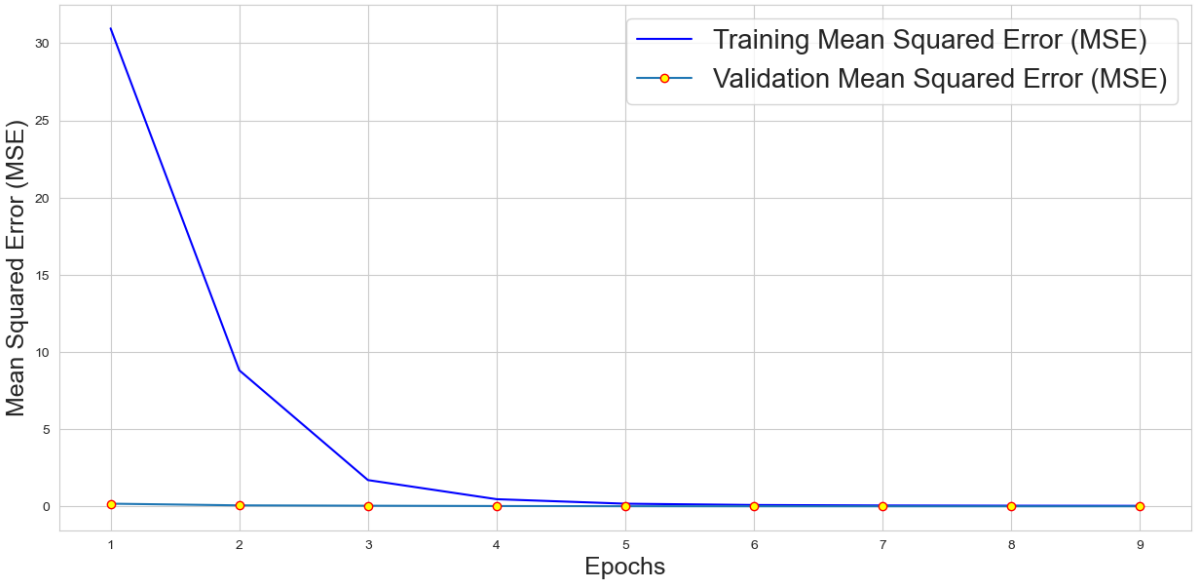
Trainable params: 1,182 (4.62 KB)

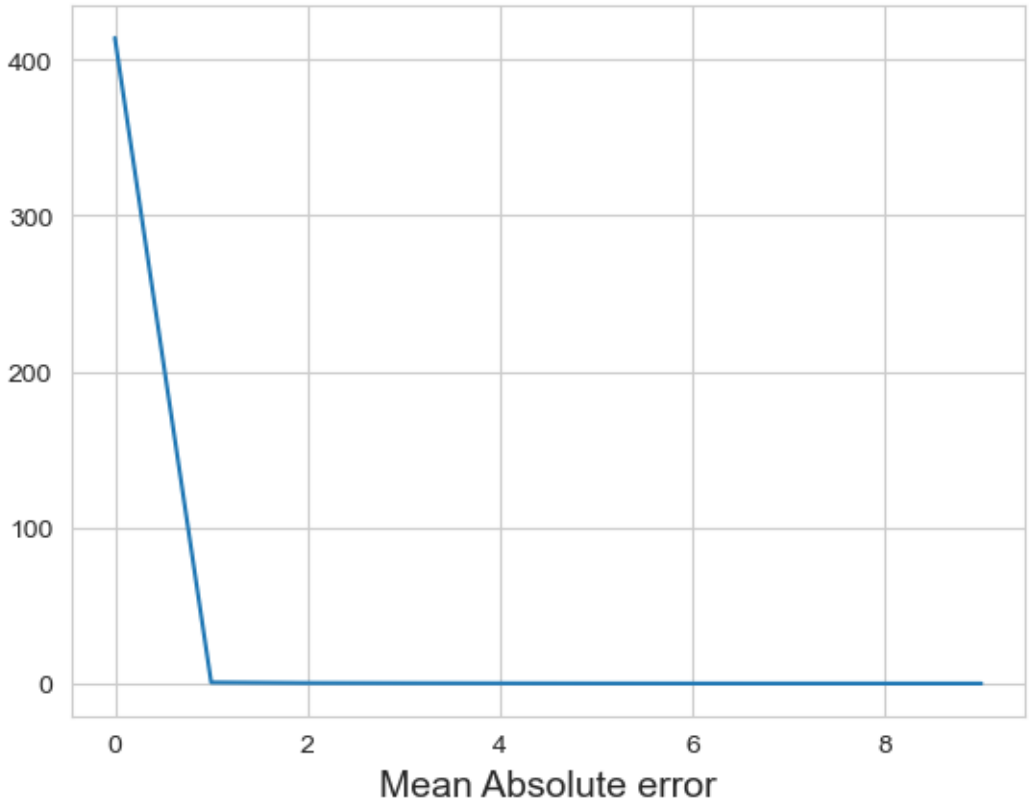
Non-trainable params: 0 (0.00 B)

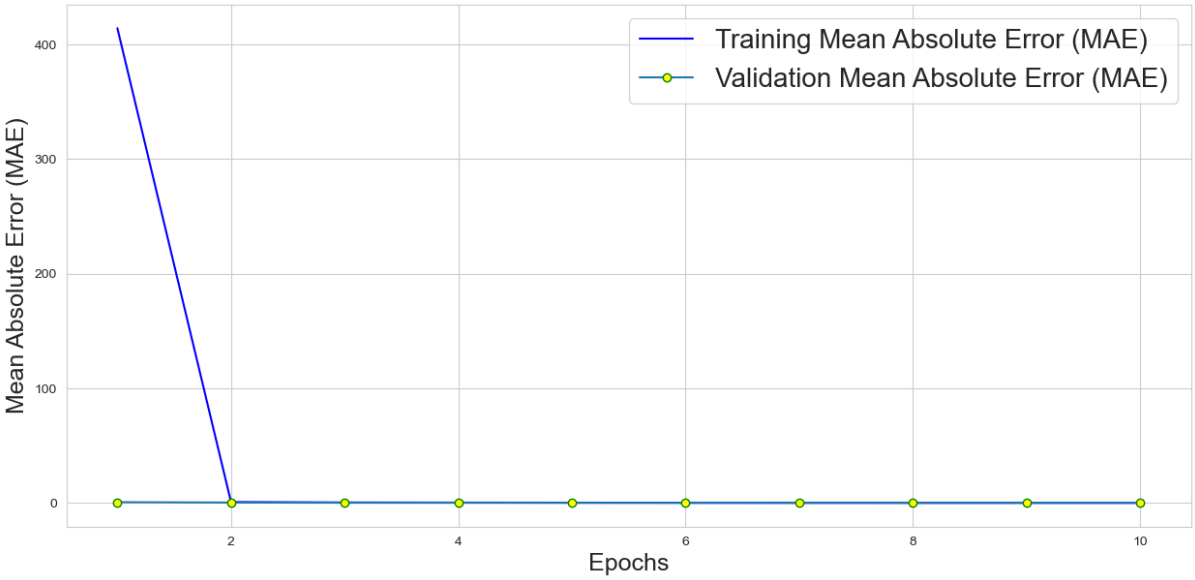
```
[91]: history = model.fit(X_train, y_train,batch_size=20,epochs=10,validation_split=0.1,verbose=1,callbacks=[es, mc])

Epoch 1/10
1421/1425 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 189308605.9550 - mae: 2539.8501
Epoch 1: val_loss improved from None to 1.52434, saving model to best_model_LSTM_Silver_3D_Tensor.keras

Epoch 1: finished saving model to best_model_LSTM_Silver_3D_Tensor.keras
1425/1425 ━━━━━━━━━━━━━━━━━━━━ 11s 7ms/step - loss: 27892186.0000 - mae: 413.9438 - val_loss: 1.5243 - val_mae: 0.6633
Epoch 2/10
1421/1425 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 43.0690 - mae: 0.9811
Epoch 2: val_loss improved from 1.52434 to 0.18010, saving model to best_model_LSTM_Silver_3D_Tensor.keras

Epoch 2: finished saving model to best_model_LSTM_Silver_3D_Tensor.keras
1425/1425 ━━━━━━━━━━━━━━━━━━━━ 9s 7ms/step - loss: 30.9651 - mae: 0.7924 - val_loss: 0.1801 - val_mae: 0.3443
Epoch 3/10
1424/1425 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.3831 - mae: 0.4310
Epoch 3: val_loss improved from 0.18010 to 0.07148, saving model to best_model_LSTM_Silver_3D_Tensor.keras

Epoch 3: finished saving model to best_model_LSTM_Silver_3D_Tensor.keras
1425/1425 ━━━━━━━━━━━━━━━━━━━━ 10s 7ms/step - loss: 8.8197 - mae: 0.4181 - val_loss: 0.0715 - val_mae: 0.1930
Epoch 4/10
1418/1425 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.2230 - mae: 0.3285
Epoch 4: val_loss improved from 0.07148 to 0.04716, saving model to best_model_LSTM_Silver_3D_Tensor.keras

Epoch 4: finished saving model to best_model_LSTM_Silver_3D_Tensor.keras
1425/1425 ━━━━━━━━━━━━━━━━━━━━ 9s 7ms/step - loss: 1.7077 - mae: 0.3126 - val_loss: 0.0472 - val_mae: 0.1556
Epoch 5/10
1420/1425 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.4851 - mae: 0.2537
Epoch 5: val_loss improved from 0.04716 to 0.02789, saving model to best_model_LSTM_Silver_3D_Tensor.keras

Epoch 5: finished saving model to best_model_LSTM_Silver_3D_Tensor.keras
1425/1425 ━━━━━━━━━━━━━━━━━━━━ 10s 7ms/step - loss: 0.4703 - mae: 0.2349 - val_loss: 0.0279 - val_mae: 0.1116
Epoch 6/10
1419/1425 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.2317 - mae: 0.1803
Epoch 6: val_loss improved from 0.02789 to 0.01564, saving model to best_model_LSTM_Silver_3D_Tensor.keras

Epoch 6: finished saving model to best_model_LSTM_Silver_3D_Tensor.keras
1425/1425 ━━━━━━━━━━━━━━━━━━━━ 10s 7ms/step - loss: 0.1788 - mae: 0.1671 - val_loss: 0.0156 - val_mae: 0.0840
Epoch 7/10
1424/1425 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.1101 - mae: 0.1333
Epoch 7: val_loss improved from 0.01564 to 0.01081, saving model to best_model_LSTM_Silver_3D_Tensor.keras

Epoch 7: finished saving model to best_model_LSTM_Silver_3D_Tensor.keras
1425/1425 ━━━━━━━━━━━━━━━━━━━━ 10s 7ms/step - loss: 0.0950 - mae: 0.1284 - val_loss: 0.0108 - val_mae: 0.0769
Epoch 8/10
1421/1425 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.0678 - mae: 0.1193
Epoch 8: val_loss improved from 0.01081 to 0.00981, saving model to best_model_LSTM_Silver_3D_Tensor.keras

Epoch 8: finished saving model to best_model_LSTM_Silver_3D_Tensor.keras
1425/1425 ━━━━━━━━━━━━━━━━━━━━ 10s 7ms/step - loss: 0.0651 - mae: 0.1177 - val_loss: 0.0098 - val_mae: 0.0729
Epoch 9/10
1418/1425 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.0537 - mae: 0.1145
Epoch 9: val_loss improved from 0.00981 to 0.00942, saving model to best_model_LSTM_Silver_3D_Tensor.keras

Epoch 9: finished saving model to best_model_LSTM_Silver_3D_Tensor.keras
1425/1425 ━━━━━━━━━━━━━━━━━━━━ 10s 7ms/step - loss: 0.0503 - mae: 0.1138 - val_loss: 0.0094 - val_mae: 0.0656
Epoch 10/10
1415/1425 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.0422 - mae: 0.1126
Epoch 10: val_loss did not improve from 0.00942
1425/1425 ━━━━━━━━━━━━━━━━━━━━ 10s 7ms/step - loss: 0.0412 - mae: 0.1106 - val_loss: 0.0108 - val_mae: 0.0692
```

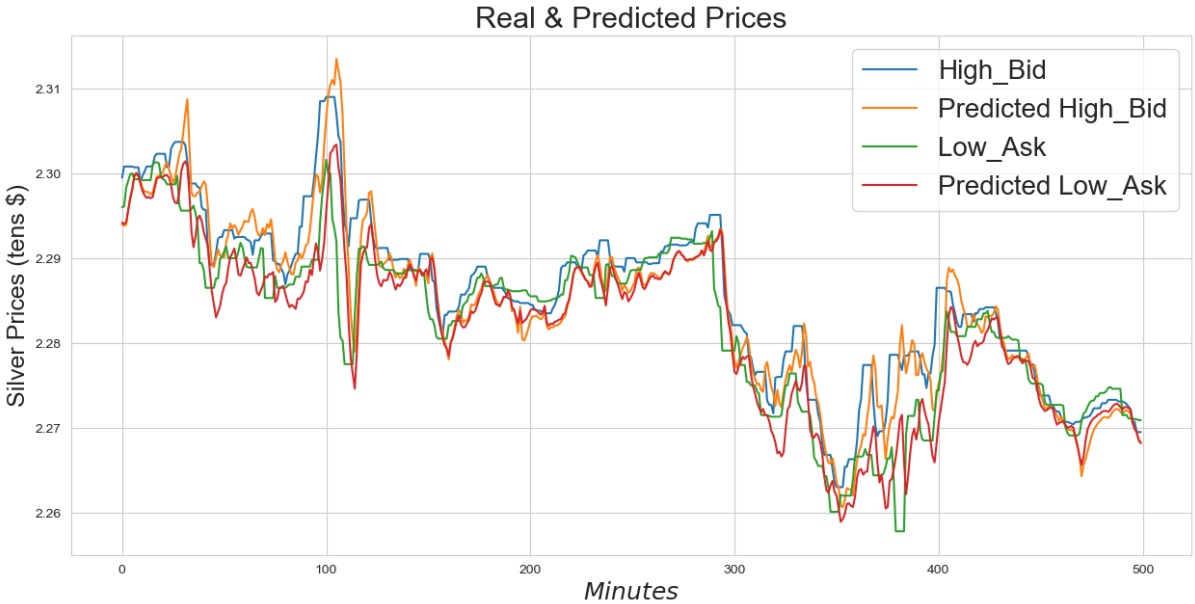

1e7
2.5
2.0
1.5
1.0
0.5
0.0
0
2
4
6
8
Mean Squared Error


1e7
Training Mean Squared Error (MSE)
Validation Mean Squared Error (MSE)
Mean Squared Error (MSE)
2.5
2.0
1.5
1.0
0.5
0.0
2
4
6
8
10
Epochs


Training Mean Squared Error (MSE)
Validation Mean Squared Error (MSE)
Mean Squared Error (MSE)
Epochs
0
5
10
15
20
25
30
1
2
3
4
5
6
7
8
9


400
300
200
100
0
0
2
4
6
8
Mean Absolute error


Training Mean Absolute Error (MAE)
Validation Mean Absolute Error (MAE)
Mean Absolute Error (MAE)
400
300
200
100
0
2
4
6
8
10
Epochs


Real & Predicted Prices
High_Bid
Predicted High_Bid
Low_Ask
Predicted Low_Ask
Silver Prices (tens $)
2.31
2.30
2.29
2.28
2.27
2.26
0
100
200
300
400
500
Minutes

WEEK -7

LOB LOGBOOK

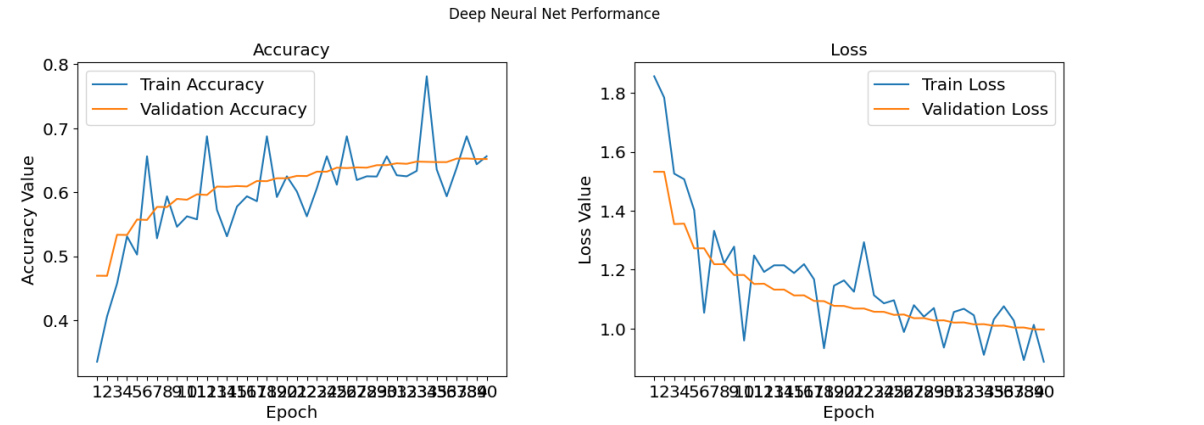
NAME - LIKHITHA MAHADEVA

SID NUMBER - 2458682

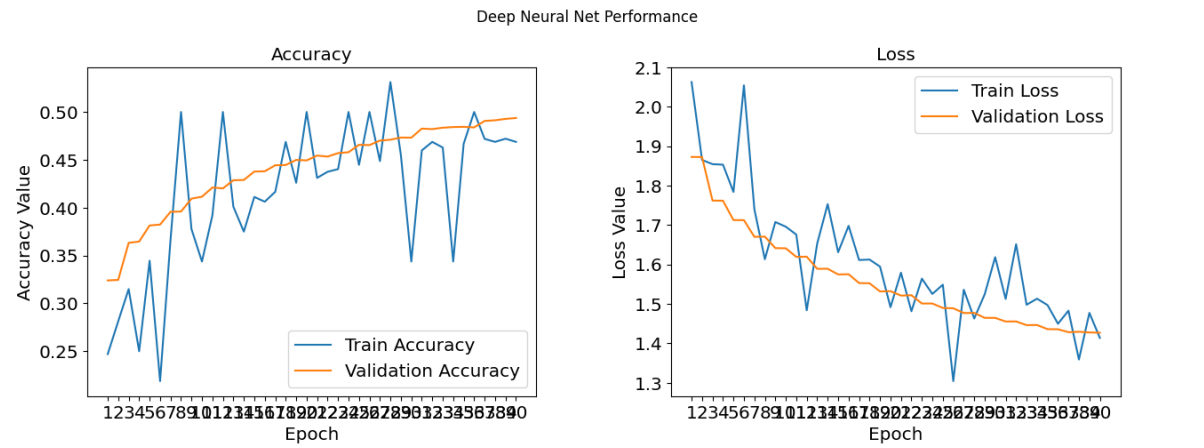
ASSIGNMENT - WEEK 7

1.Graphs for Accuracy and Loss Function for VGG16 and VGG19 (4 graphs in total).

```
ax2.plot(epochs, history.history['loss'], label='Train Loss')
ax2.plot(epochs, history.history['val_loss'], label='Validation Loss')
ax2.set_xticks(epochs)
ax2.set_ylabel('Loss Value')
ax2.set_xlabel('Epoch')
ax2.set_title('Loss')
l2 = ax2.legend(loc="best")
```




```
ax2.plot(epochs, history_2.history['loss'], label='Train Loss')
ax2.plot(epochs, history_2.history['val_loss'], label='Validation Loss')
ax2.set_xticks(epochs)
ax2.set_ylabel('Loss Value')
ax2.set_xlabel('Epoch')
ax2.set_title('Loss')
l2 = ax2.legend(loc="best")
```



the optimal number of epochs for training your VGG16 and VGG19 models.

early-stop code

```
from tensorflow.keras.callbacks import EarlyStopping
LAST_DIGIT_IN_YOUR_SID = 12

early_stop =
EarlyStopping(monitor='val_loss',patience=LAST_DIGIT_IN_YOUR_SID)
```

Add the final Performance Table

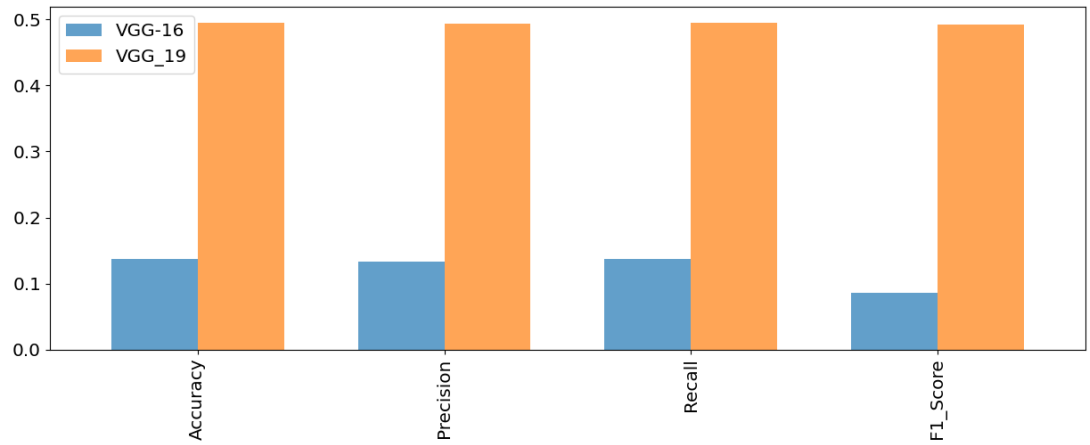
```
ErrorMetrics_pd.columns = ['Accuracy', 'Precision', 'Recall', 'F1_Score']
ErrorMetrics_pd
```

	Accuracy	Precision	Recall	F1_Score
VGG-16	0.1371	0.133286	0.1371	0.085890
VGG_19	0.4951	0.493726	0.4951	0.491722

Plot the performance comparing histogram of VGG16 and VGG19

```
ErrorMetrics_pd.plot.bar(width=0.7,alpha=0.7)
```

```
<Axes: >
```



WEEK -6

LOB LOGBOOK

NAME - LIKHITHA MAHADEVA

SID NUMBER - 2458682

ASSIGNMENT - WEEK 6

[78]:
```
# Use any neural network model based on RNN
# You can also use the model created above
#
# Create the model

# ...
# ...
# ...
model = Sequential()

model.add(SimpleRNN(130, activation='relu',
                return_sequences=True,
                input_shape=(length, n_features)))

model.add(SimpleRNN(170, activation='relu'))

model.add(Dense(1))
```

Model: "sequential_5"

Layer (type)	Output Shape	Param #
simple_rnn (SimpleRNN)	(None, 12, 130)	17,160
simple_rnn_1 (SimpleRNN)	(None, 170)	51,170
dense_2 (Dense)	(None, 1)	171

Total params: 68,501 (267.58 KB)

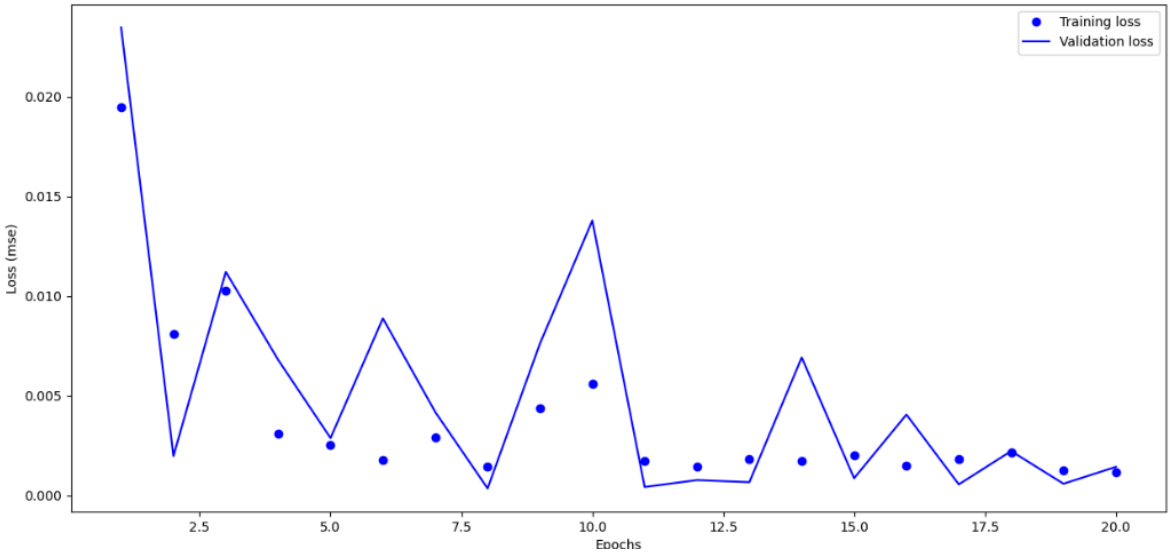
Trainable params: 68,501 (267.58 KB)

Non-trainable params: 0 (0.00 B)

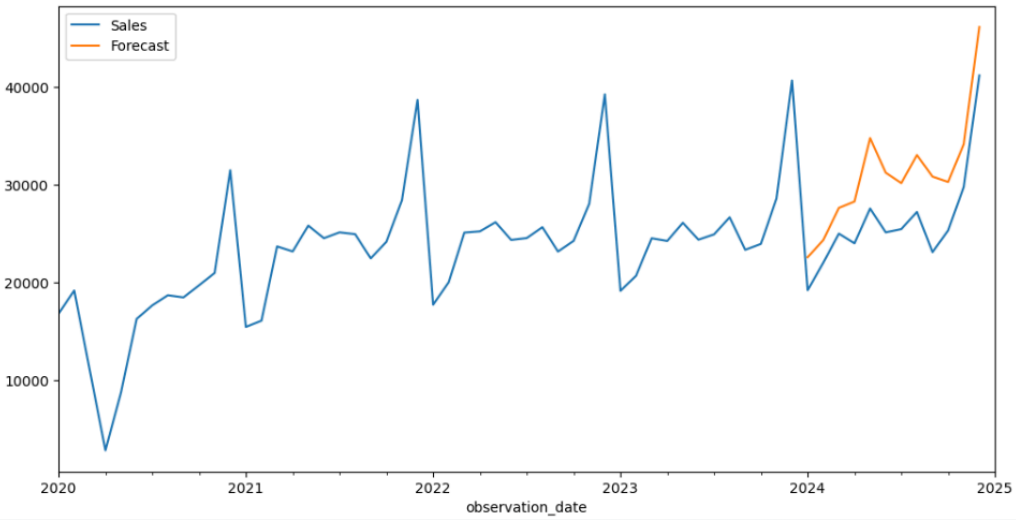
```python
# Your code to create an object early-stop

# Use the patience parameter equal to the last digit in your SID.
# However,
# If the last digit of your SID = 0, then assign patience=10
# If the last digit of your SID = 1, then assign patience=11
# If the last digit of your SID = 2, then assign patience=12
# If the last digit of your SID = 3, then assign patience=13

# For example, your SID = 22984879, then patience=9
# One more example: your SID = 22984883, then patience=13

LAST_DIGIT_IN_YOUR_SID = 12

early_stop = EarlyStopping(monitor='val_loss',patience=LAST_DIGIT_IN_YOUR_SID)
```

```
Epoch 1/30
384/384 ━━━━━━━━━━━━━━━━━━━━ 4s 4ms/step - loss: 0.0195 - val_loss: 0.0235
Epoch 2/30
384/384 ━━━━━━━━━━━━━━━━━━━━ 2s 3ms/step - loss: 0.0081 - val_loss: 0.0020
Epoch 3/30
384/384 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0103 - val_loss: 0.0112
Epoch 4/30
384/384 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0031 - val_loss: 0.0068
Epoch 5/30
384/384 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0025 - val_loss: 0.0029
Epoch 6/30
384/384 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0018 - val_loss: 0.0089
Epoch 7/30
384/384 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0029 - val_loss: 0.0042
Epoch 8/30
384/384 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0014 - val_loss: 3.4910e-04
Epoch 9/30
384/384 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0044 - val_loss: 0.0076
Epoch 10/30
384/384 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0056 - val_loss: 0.0138
Epoch 11/30
384/384 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0017 - val_loss: 4.2040e-04
Epoch 12/30
384/384 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0014 - val_loss: 7.6541e-04
Epoch 13/30
384/384 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0018 - val_loss: 6.6156e-04
Epoch 14/30
384/384 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0017 - val_loss: 0.0069
Epoch 15/30
384/384 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0020 - val_loss: 8.6020e-04
Epoch 16/30
384/384 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0015 - val_loss: 0.0041
Epoch 17/30
384/384 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0018 - val_loss: 5.5051e-04
Epoch 18/30
384/384 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0022 - val_loss: 0.0022
Epoch 19/30
384/384 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0013 - val_loss: 5.8301e-04
Epoch 20/30
384/384 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 0.0012 - val_loss: 0.0014
```

```
plt.plot(epochs, val_acc_values, 'b', label='Validation loss')
plt.xlabel('Epochs')
plt.ylabel('Loss (mse)')
plt.legend()

plt.show()
```



[104]:
```
ax=df.plot(figsize=(12,6))
forecast_df.plot(ax=ax)
plt.xlim('2020-01-01','2025-01-01')
plt.show();
```



WEEK -5

LOB LOGBOOK

NAME - LIKHITHA MAHADEVA

SID NUMBER - 2458682

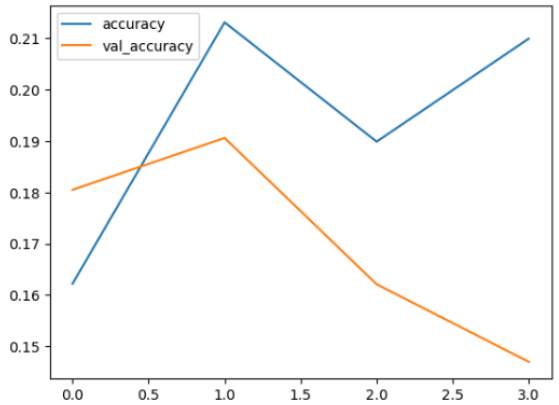
ASSIGNMENT - WEEK 5

Code & graphs :

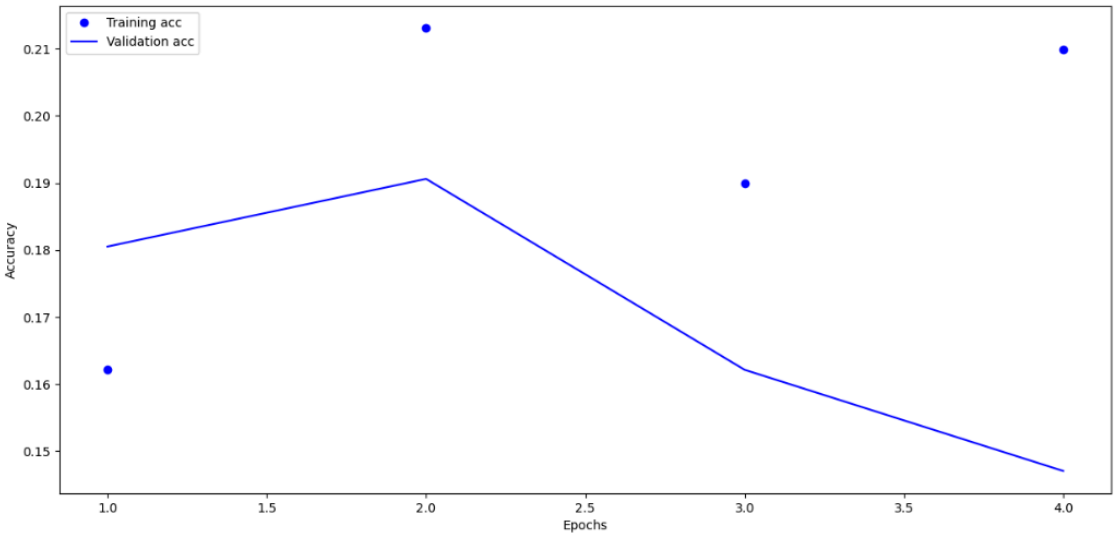
```
# Plot accuracy and val_accuracy vs the epochs

losses[['accuracy','val_accuracy']].plot()
```

```
[39]: losses[['accuracy','val_accuracy']].plot()
```

```
[39]: <Axes: >
```

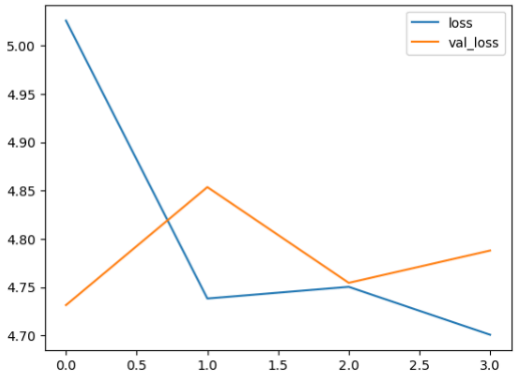



```
history_dict = history.history

acc_values = history_dict['accuracy']
val_acc_values = history_dict['val_accuracy']
epochs = range(1, len(acc_values) + 1)

plt.figure(num=1, figsize=(15,7))
plt.plot(epochs, acc_values, 'bo', label='Training acc')
plt.plot(epochs, val_acc_values, 'b', label='Validation acc')
plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.legend()

plt.show()
```



```
[41]: losses[['loss','val_loss']].plot()
```

```
[41]: <Axes: >
```




```
[26]: model=Sequential()
model.add(Conv2D(filters=32,kernel_size=(4,4),strides=(1,1),input_shape=(32,32,3),activation='relu',))
model.add(MaxPool2D(pool_size=(2,2),strides=(2,2)))

model.add(Conv2D(filters=64,kernel_size=(3,3),strides=(2,2),activation='relu',))
model.add(MaxPool2D(pool_size=(2,2),strides=(2,2)))


model.add(Flatten())


model.add(Dense(128,activation='relu'))
model.add(Dense(10,activation='relu'))
```

```
[ ]:
```

Model summary:

```
[27]: model.summary()
```

Model: "sequential_1"

Layer (type)	Output Shape	Param #
conv2d_1 (Conv2D)	(None, 29, 29, 32)	1,568
max_pooling2d_1 (MaxPooling2D)	(None, 14, 14, 32)	0
conv2d_2 (Conv2D)	(None, 6, 6, 64)	18,496
max_pooling2d_2 (MaxPooling2D)	(None, 3, 3, 64)	0
flatten_1 (Flatten)	(None, 576)	0
dense (Dense)	(None, 128)	73,856
dense_1 (Dense)	(None, 10)	1,290

Total params: 95,210 (371.91 KB)
Trainable params: 95,210 (371.91 KB)
Non-trainable params: 0 (0.00 B)

WEEK -4

LOB LOGBOOK

NAME - LIKHITHA MAHADEVA

SID NUMBER - 2458682

ASSIGNMENT - WEEK 4

Model summary

Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 152)	3,040
dense_1 (Dense)	(None, 130)	19,890
dense_2 (Dense)	(None, 112)	14,672
dense_3 (Dense)	(None, 96)	10,848
dense_4 (Dense)	(None, 68)	6,596
dense_5 (Dense)	(None, 40)	2,760
dense_6 (Dense)	(None, 1)	41

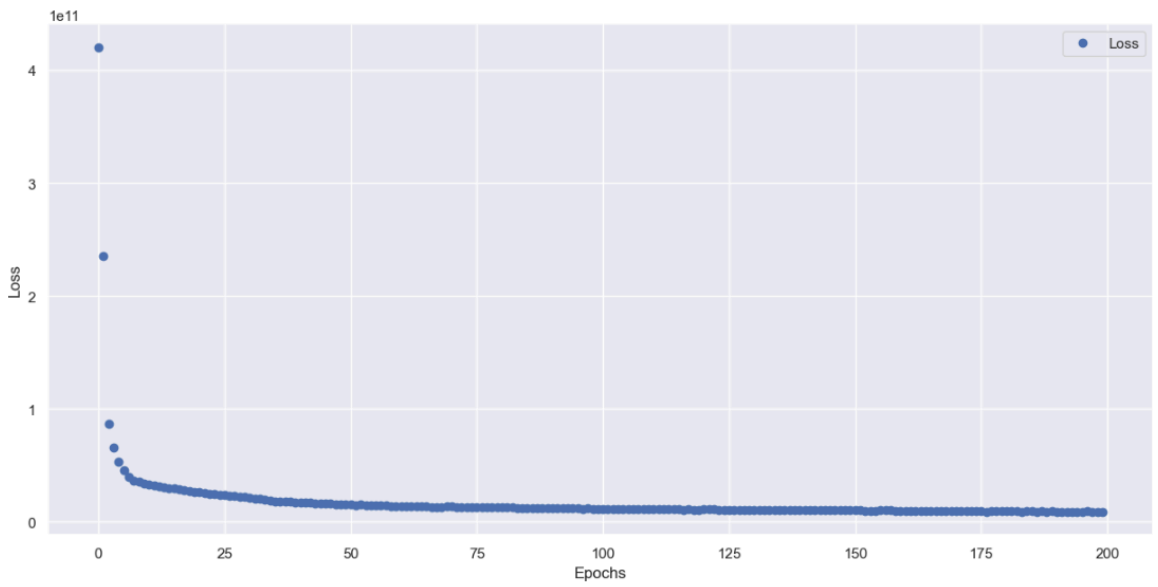
Total params: 115,696 (451.94 KB)

Trainable params: 57,847 (225.96 KB)

Non-trainable params: 0 (0.00 B)

Optimizer params: 57,849 (225.98 KB)

Loss after every epoch :



WEEK -3

LOB LOGBOOK

NAME - LIKHITHA MAHADEVA

SID NUMBER - 2458682

ASSIGNMENT - WEEK 3

Accuracy :

```
[45]: test_loss, test_acc = model.evaluate(test_images, test_labels, verbose=2)
      print('\nTest accuracy:', test_acc)
```

```
313/313 - 9s - 27ms/step - accuracy: 0.8740 - loss: 0.5104
```

Code :

```
[44]: print(model.summary(line_length=None,
          positions=None,
          print_fn=None,
          expand_nested=False,
          show_trainable=False,
          layer_range=None,))
```

Model: "sequential"

Layer (type)	Output Shape	Param #
flatten (Flatten)	(None, 784)	0
dense (Dense)	(None, 150)	117,750
dense_1 (Dense)	(None, 20)	3,020
dense_2 (Dense)	(None, 180)	3,780
dense_3 (Dense)	(None, 10)	1,810

Total params: 379,082 (1.45 MB)

Trainable params: 126,360 (493.59 KB)

Non-trainable params: 0 (0.00 B)

Optimizer params: 252,722 (987.20 KB)

None

```
[21]: # The result will be different, as your particular unique model should be demonstrated here
```

Sum:

$w_1, w_2, w_3, w_4, w_5, w_6, w_7, w_8 = 0.2$

$w_9 = w_{10} = w_{11} = w_{12} = 0.1$

$w_{13} = w_{14} = w_{15} = w_{16} = 0.3$.

New var:

$y = \frac{SSID}{8964879}$

$Z = b + \sum_{i=1}^{m} w_i x_i$

inputs

$b = 0$.

(3) $Z_1 = Z_2 = 0.2(x_1 + x_2 + x_3 + x_4)$

$= 0.2 \cdot y(23 + 32 + 56 + 48)$

$= 0.2 \cdot 159 \cdot y = \underline{31.8 \cdot y}$

(4) $y > 0 \Rightarrow h_1 = h_2 = \text{ReLU}$

$\begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix} = 31.8 \cdot y$.

(5) $Z_3 = Z_4 = 0.1(h_1 + h_2) = 0.1 \cdot 2 \cdot 31.8 \cdot y = 6.36 \cdot y$

(6) $y > 0 \Rightarrow h_3 = h_4 = \text{ReLU}\begin{pmatrix} Z_3 \\ Z_4 \end{pmatrix} = 6.36 \cdot y$

(7) $0.3(h_3 + h_4) = 0.3 \cdot 2 \cdot 6.36 \cdot y = 3.816 \cdot y$

(8) $Z_5 = Z_6 = 3.816 \cdot y + 0.5$

(9) $O_1 = O_2 = \sigma\begin{pmatrix} Z_5 \\ Z_6 \end{pmatrix} = \sigma(3.816 \cdot y + 0.5)$

$= \sigma\left(3{,}816 \cdot \frac{SSID}{8964879} + 0.5\right)$

$SSID = 2373000$

$\Rightarrow O_1 = O_2 = \sigma\left(3.816 \cdot \frac{2373000}{8964879} + 0.5\right)$

$= \sigma(1.5100937)$

$\cong \underline{0{,}819}$

WEEK -2

LOB LOGBOOK

NAME - LIKHITHA MAHADEVA

SID NUMBER - 2458682

ASSIGNMENT - WEEK 1

Accuracy :0.845

Code :

```
[23]: print(model.summary())
```

```
Model: "sequential"
```

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 10)	350
dense_1 (Dense)	(None, 8)	88
dense_2 (Dense)	(None, 1)	9

```
Total params: 447 (1.75 KB)
Trainable params: 447 (1.75 KB)
Non-trainable params: 0 (0.00 B)
None
```

The architecture table will differ from the one below and will depend on the number of layers and cells you selected above. However, the overall appearance of the table will be similar.

- x1=6.3079
- x2=8.7762
- x3=15.3584
- x4=13.1643

All input weights = **0.2**

h1=h2=8.7214

Weights = **0.1**

h3=h4=1.7443

Bias = **1 × 0.5**

Weights = **0.3**

O1=0.8244

O2=0.8244

Sum:

$w_1, w_2, w_3, w_4, w_5, w_6, w_7, w_8 = 0.2$

$w_9 = w_{10} = w_{11} = w_{12} = 0.1$

$w_{13} = w_{14} = w_{15} = w_{16} = 0.3$.

New var:

$y = \frac{SSID}{8964879}$

$Z = b + \sum_{i=1}^{m} w_i x_i$

inputs

$b = 0$.

(3) $Z_1 = Z_2 = 0.2(x_1 + x_2 + x_3 + x_4)$

$= 0.2 \cdot y(23 + 32 + 56 + 48)$

$= 0.2 \cdot 159 \cdot y = \underline{31.8 \cdot y}$

(4) $y > 0 \Rightarrow h_1 = h_2 = \text{ReLU}$

$\begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix} = 31.8 \cdot y$.

(5) $Z_3 = Z_4 = 0.1(h_1 + h_2) = 0.1 \cdot 2 \cdot 31.8 \cdot y = 6.36 \cdot y$

(6) $y > 0 \Rightarrow h_3 = h_4 = \text{ReLU}\begin{pmatrix} Z_3 \\ Z_4 \end{pmatrix} = 6.36 \cdot y$

(7) $0.3(h_3 + h_4) = 0.3 \cdot 2 \cdot 6.36 \cdot y = 3.816 \cdot y$

(8) $Z_5 = Z_6 = 3.816 \cdot y + 0.5$

(9) $O_1 = O_2 = \sigma\begin{pmatrix} Z_5 \\ Z_6 \end{pmatrix} = \sigma(3.816 \cdot y + 0.5)$

$= \sigma\left(3{,}816 \cdot \frac{SSID}{8964879} + 0.5\right)$

$SSID = 2373000$

$\Rightarrow O_1 = O_2 = \sigma\left(3.816 \cdot \frac{2373000}{8964879} + 0.5\right)$

$= \sigma(1.5100937)$

$\cong \underline{0{,}819}$

NAME - LIKHITHA MAHADEVA

SID NUMBER - 2458682

ASSIGNMENT - WEEK 1

CODE -

jupyter Week1-LinearRegression_tensorflow-St_2026 Last Checkpoint: 37 minutes ago

File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python [conda env:base] *

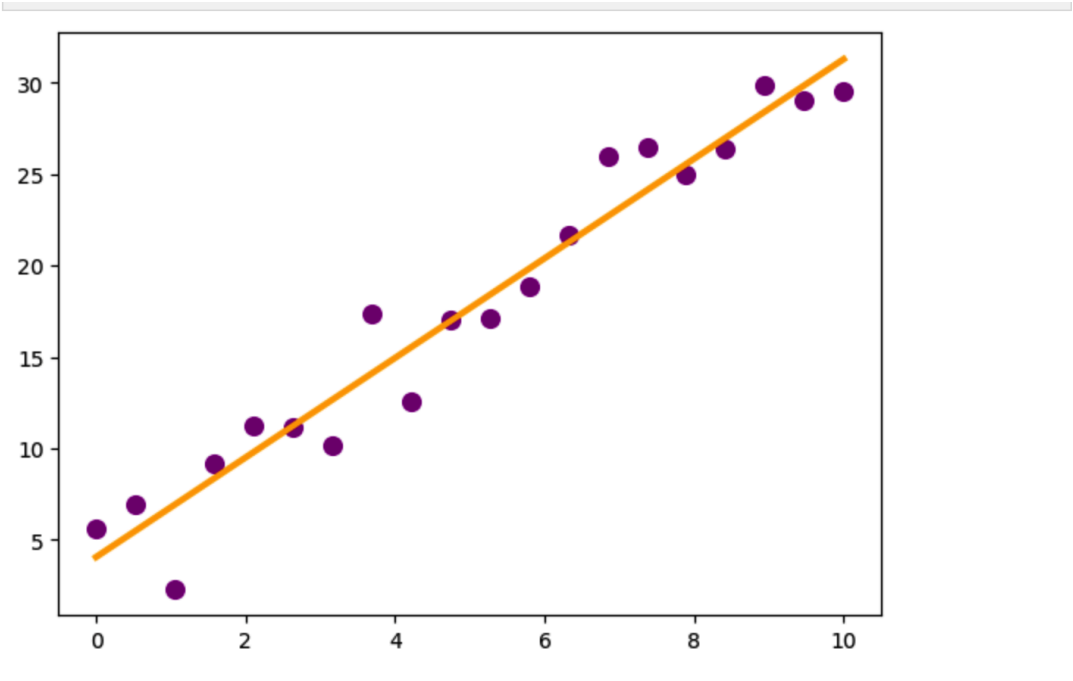
Lab Logbook requirements:

Please make sure to document the following in your Lablogbook:

Include an image of the generated graph depicting the original data and the fitted line for linear regression. Modify the appearance of the graph by adjusting its color or size, making it different from the graph of your classmates. Copy the figure and paste it directly into your logbook. Refrain from taking a snapshot; instead, right-click and copy the graph to your lab logbook. Ensure that no code or other information is added to the logbook and that only required graph is present.

Marks will not be awarded if anything else is found in the logbook or instructions are not clearly followed.

[49]:

```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(0, 10, 20)
y = 2.5 * x + 5 + np.random.normal(0, 3, size=len(x))

coeffs = np.polyfit(x, y, 1)
y_fit = np.polyval(coeffs, x)

plt.figure(figsize=(7, 5))
plt.scatter(x, y, color='purple', s=70)
plt.plot(x, y_fit, color='orange', linewidth=3)
plt.show()
```

OUTPUT -